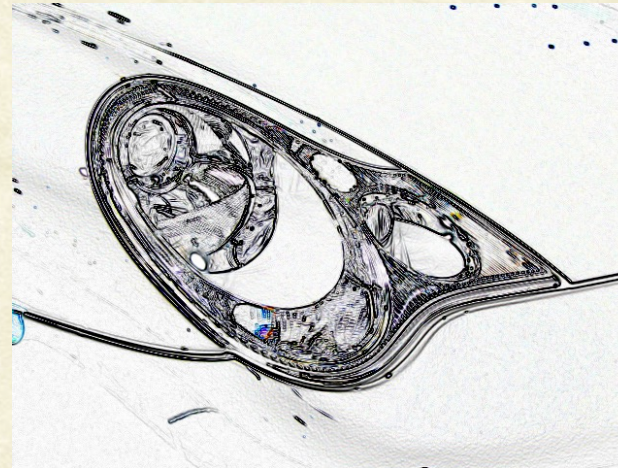




CS7.404: Digital Image Processing

Monsoon 2025: Image Segmentation - 2



Anoop M. Namboodiri
Biometrics and Secure ID Lab, CVIT,
IIIT Hyderabad

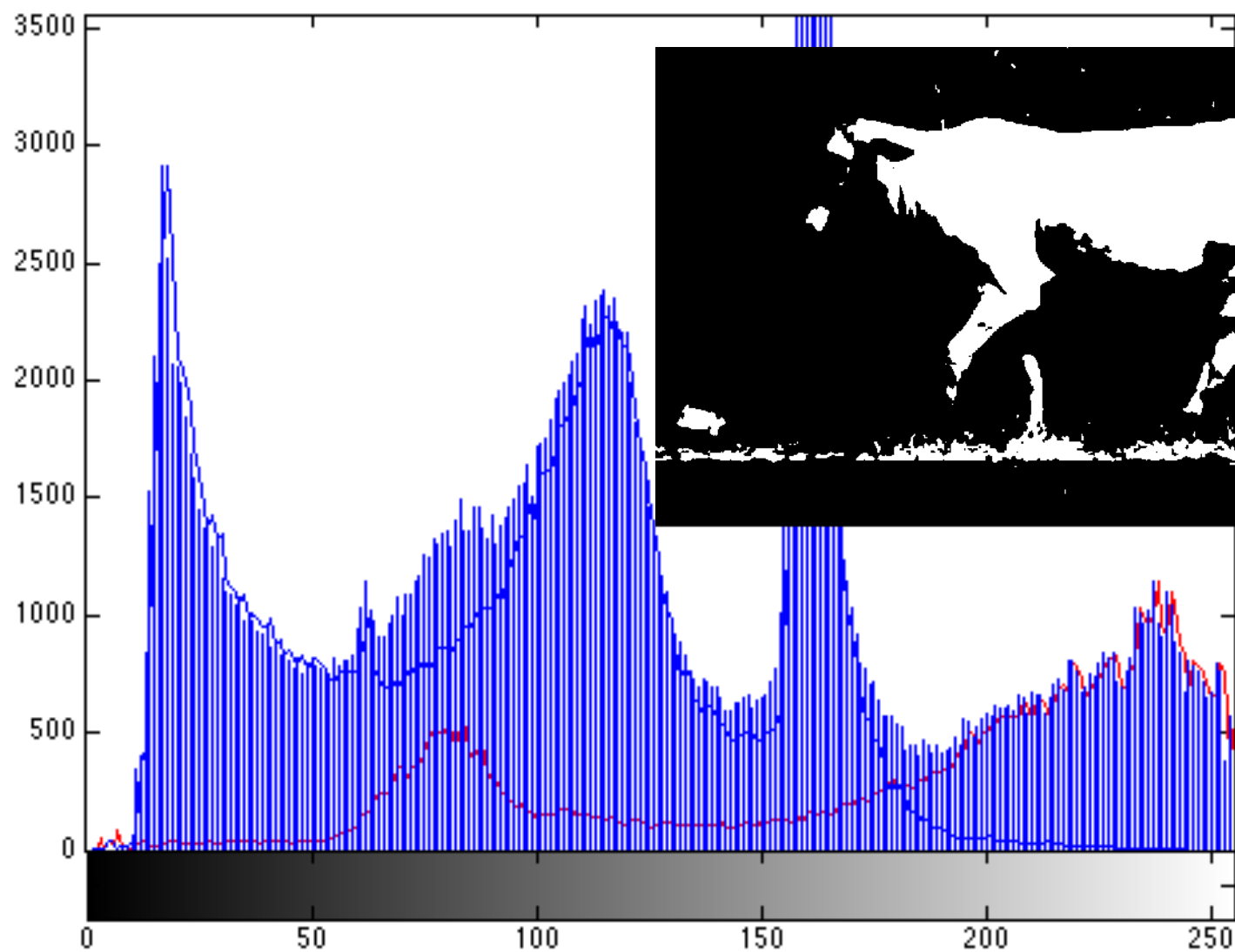
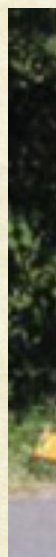


Types of Segmentation

- Classification-based
 - Label pixels based on region properties
 - Label each pixel based on object models
- Region-based
 - Region growing and splitting
- Boundary-based
 - Find edges in the image and use them as region boundary
- Motion-based
 - Group pixels that have consistent motion (e.g., move in the same direction)



Is It





Segmentation by Pixel Classification

Two Primary Challenges:

1. How to use object / background properties to decide on pixel label?
 - e.g., Ducks are white and yellow, while background is green and brown
2. How to ensure that regions are continuous regions?
 - Avoid fragmentation of object regions

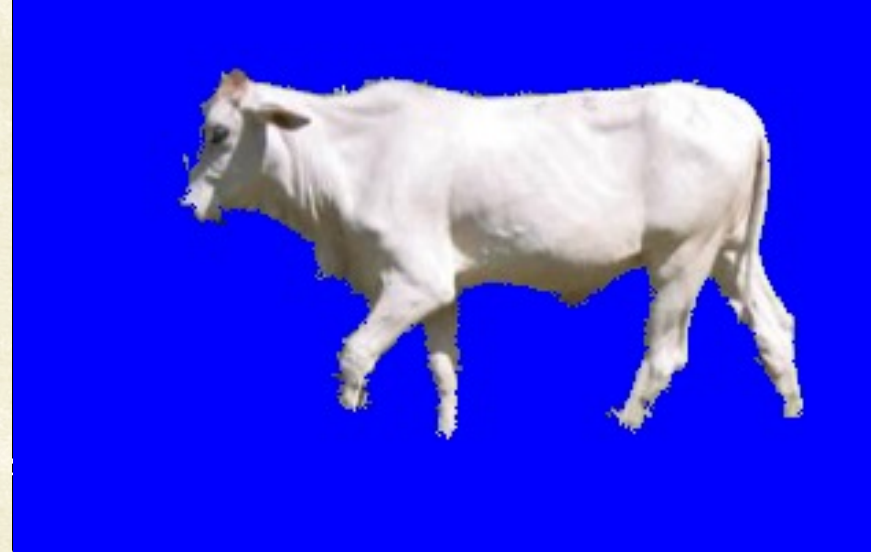


Segmentation as Optimal Labeling

- Model knowledge about the world
- Classify each pixel as belonging to a specific object
 - Independent classification does not work
 - Need to incorporate neighborhood information
- Consider a graph over the image
 - Each node in the graph need to be labeled
 - Edges in the graph represent neighborhood constraints
- Define a cost function, $Q(f)$, using the above
- Compute the optimal labeling wrt $Q(f)$.



Binary Image Segmentation



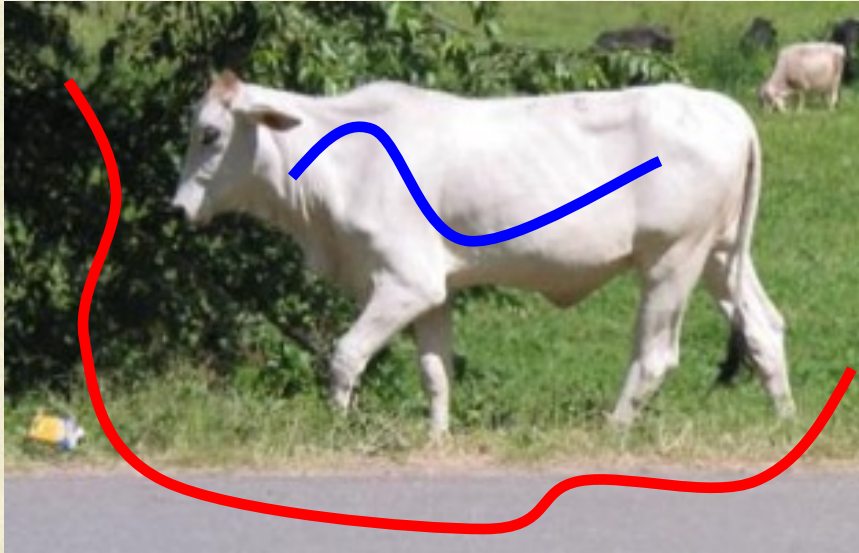
How ?

Cost function Models *our* knowledge about natural images

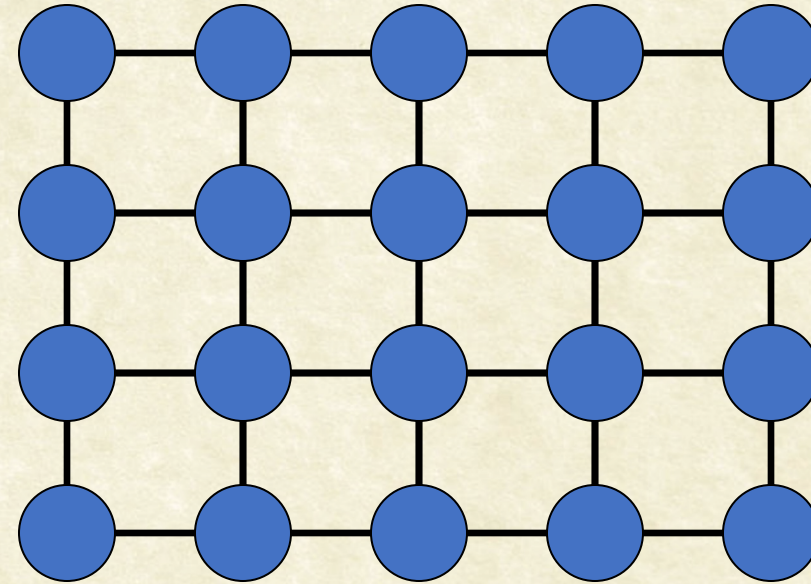
Optimize cost function to obtain the segmentation



Binary Image Segmentation



Object - white, Background - green/grey



Graph $G = (V, E)$

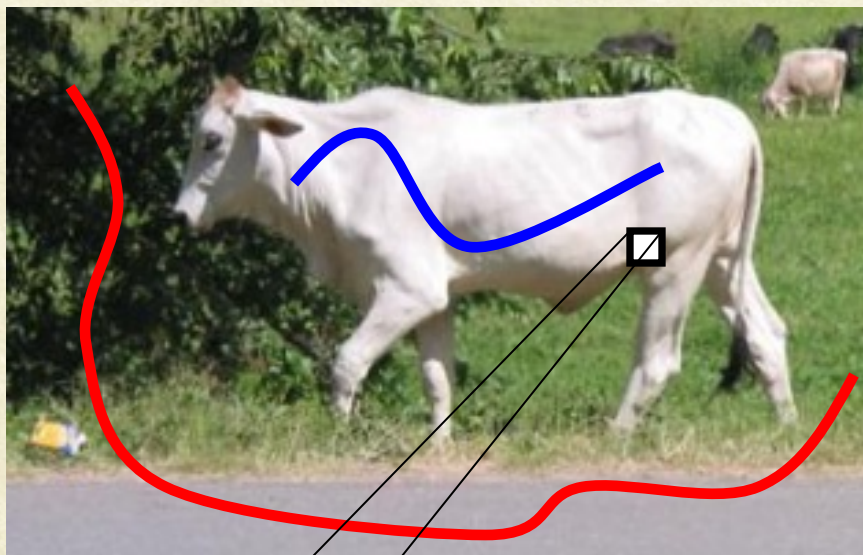
Each vertex corresponds to a pixel

Edges define a 4-neighbourhood *grid* graph

Assign a label to each vertex from $L = \{\text{obj}, \text{bkg}\}$



Binary Image Segmentation

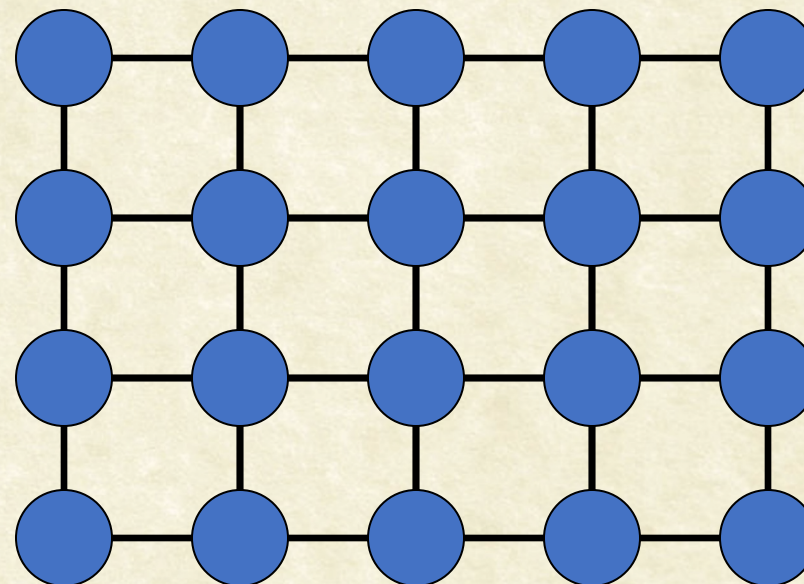


Object - white, Background - green/grey

Cost of a labelling $f : V \rightarrow L$



Cost of label 'obj' low Cost of label 'bkg' high

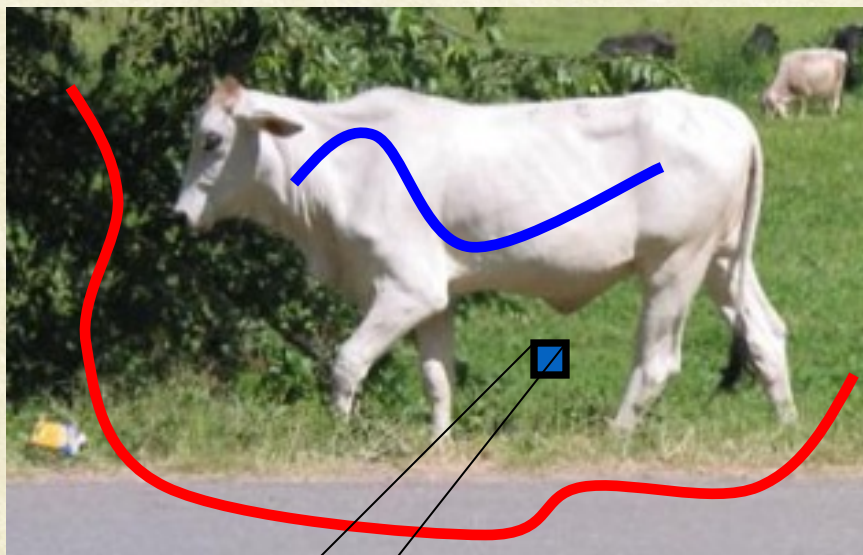


Graph $G = (V, E)$

Per Vertex Cost



Binary Image Segmentation



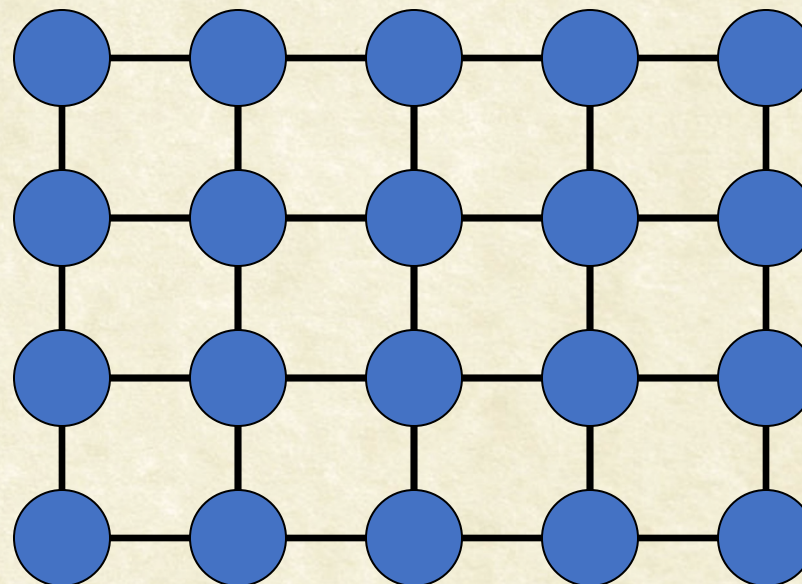
Object - white, Background - green/grey

Cost of a labelling $f : V \rightarrow L$



Cost of label 'obj' high Cost of label 'bkg' low

UNARY COST

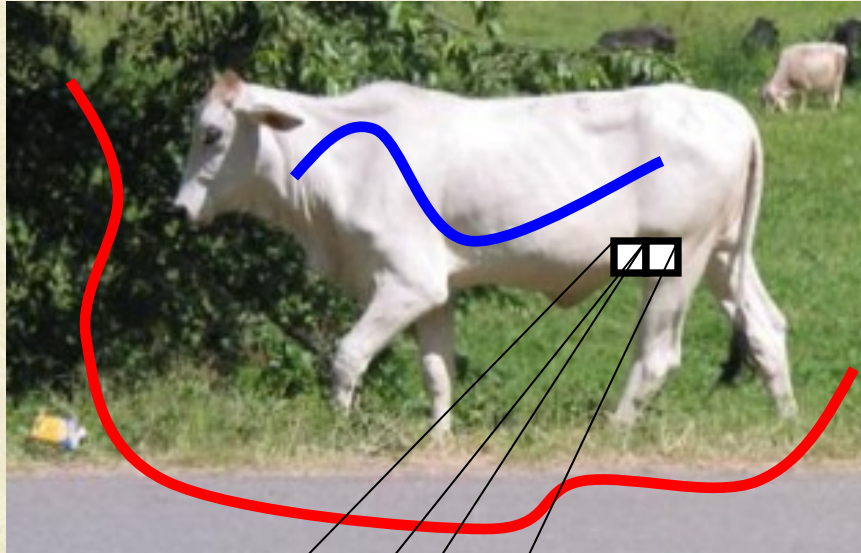


Graph $G = (V, E)$

Per Vertex Cost



Binary Image Segmentation



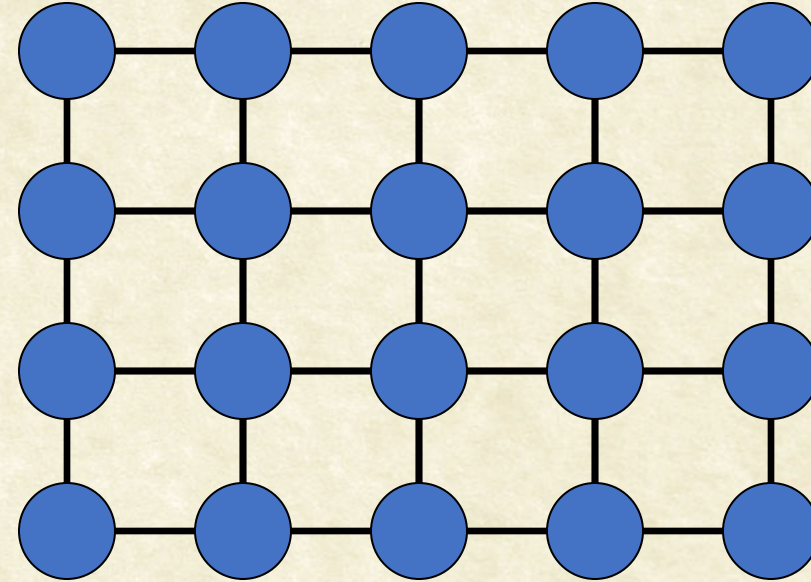
Object - white, Background - green/grey

Cost of a labelling $f : V \rightarrow L$



Cost of same label low

Cost of different labels high

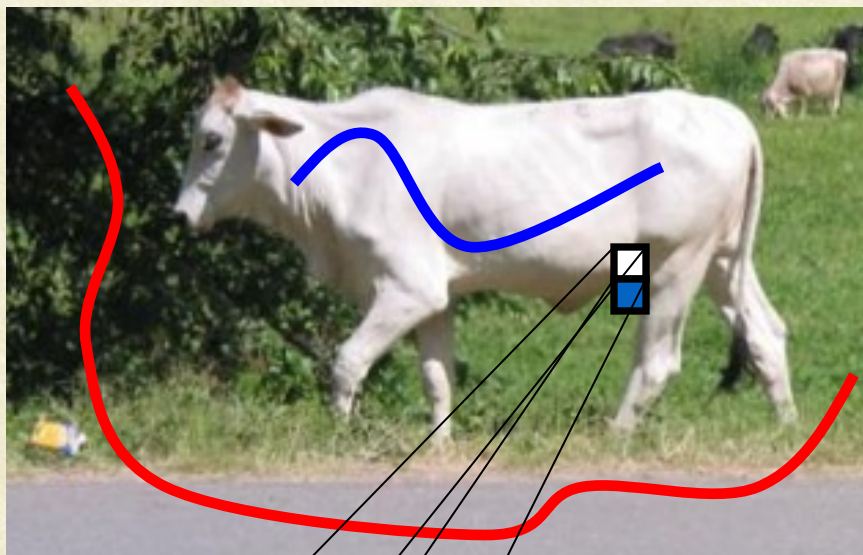


Graph $G = (V, E)$

Per Edge Cost

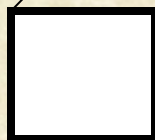


Binary Image Segmentation



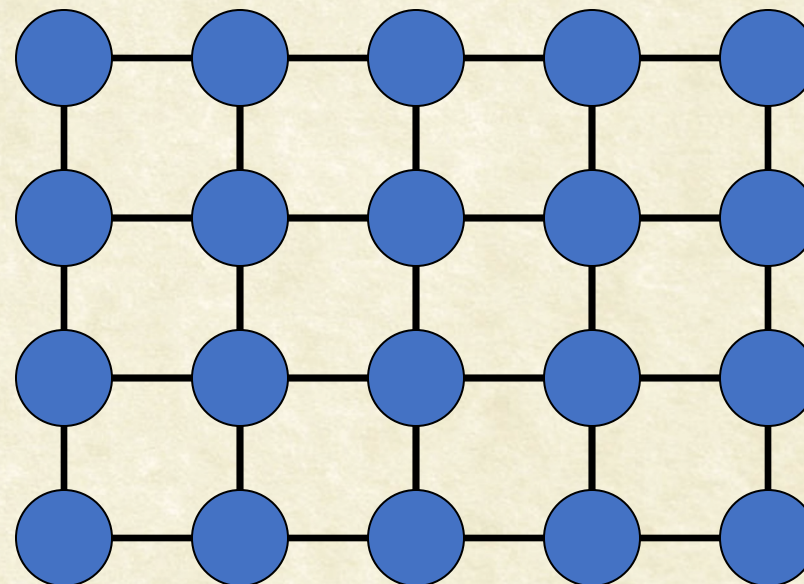
Object - white, Background - green/grey

Cost of a labelling $f : V \rightarrow L$



Cost of same label high

Cost of different labels low



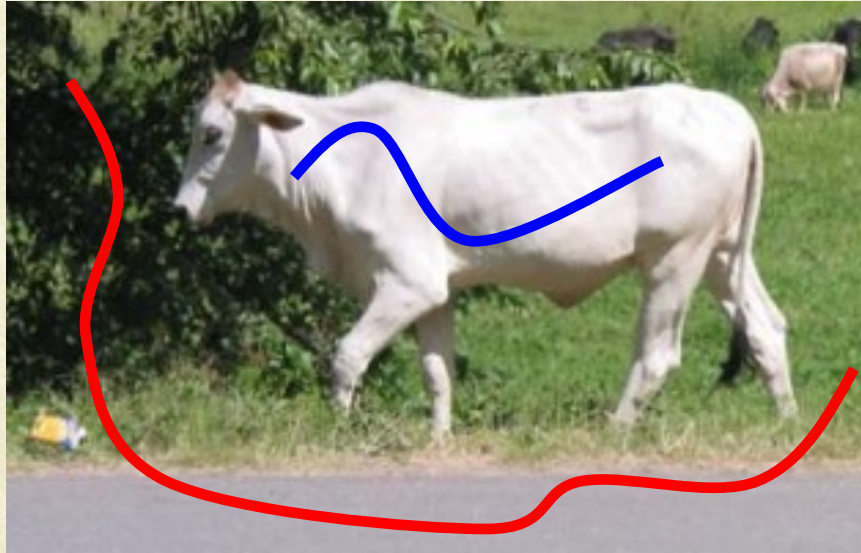
Graph $G = (V, E)$

Per Edge Cost

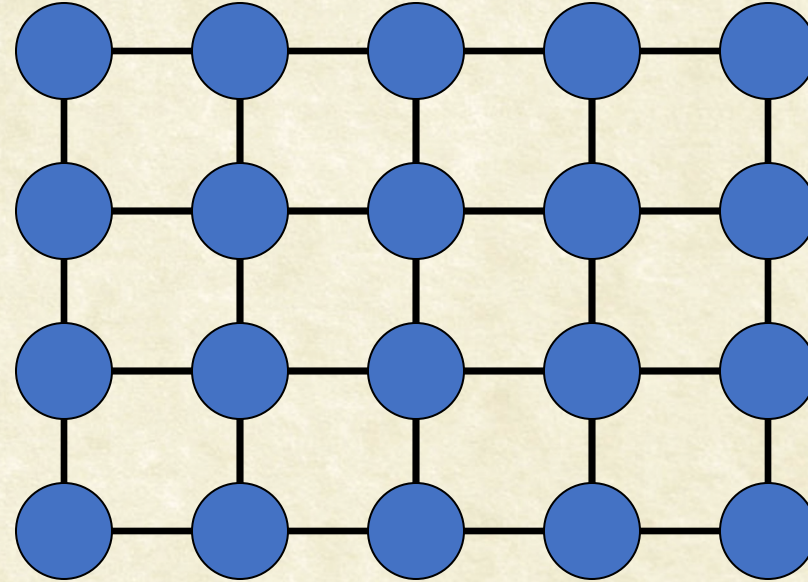
PAIRWISE
COST



Binary Image Segmentation



Object - white, Background - green/grey

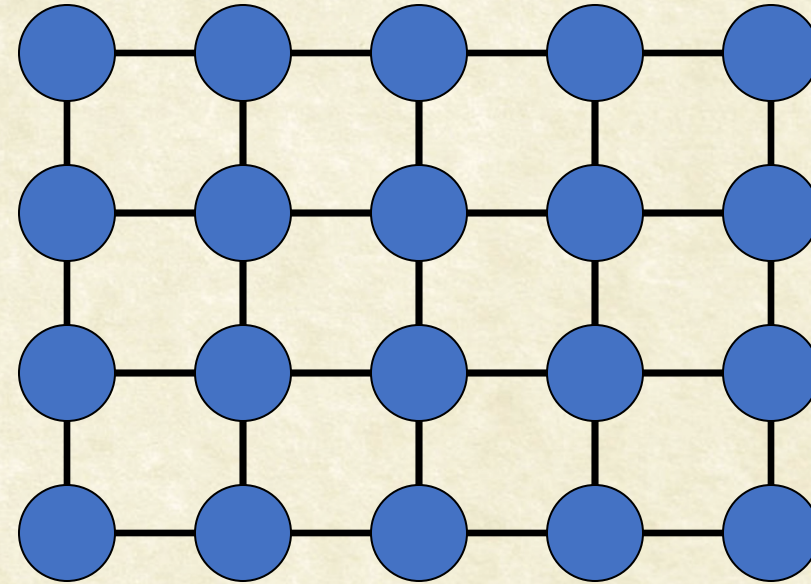
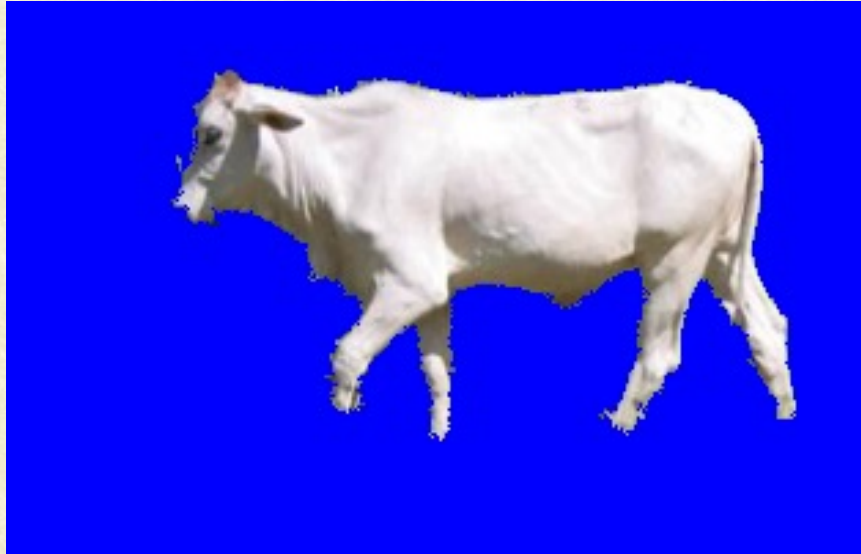


Graph $G = (V, E)$

Problem: Find the labeling with minimum cost f^*



Binary Image Segmentation



Graph $G = (V, E)$

Vertex corresponds to a pixel

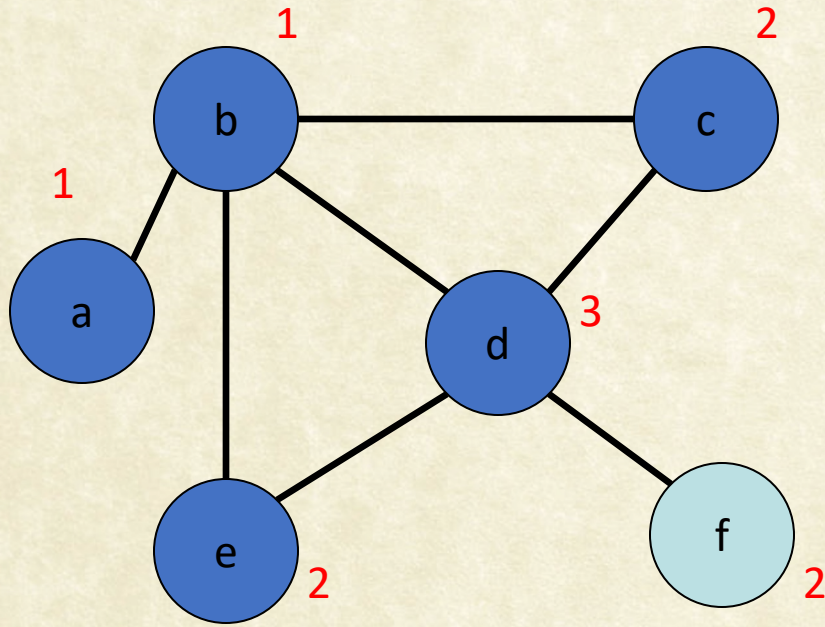
Edges define grid graph

$$L = \{fg, bg\}$$

Problem: Find the labeling with minimum cost f^*



The General Problem



Graph $G = (V, E)$

Discrete label set $L = \{1, 2, \dots, h\}$

Assign a label to each vertex

$f: V \rightarrow L$

Cost of a labelling $Q(f)$

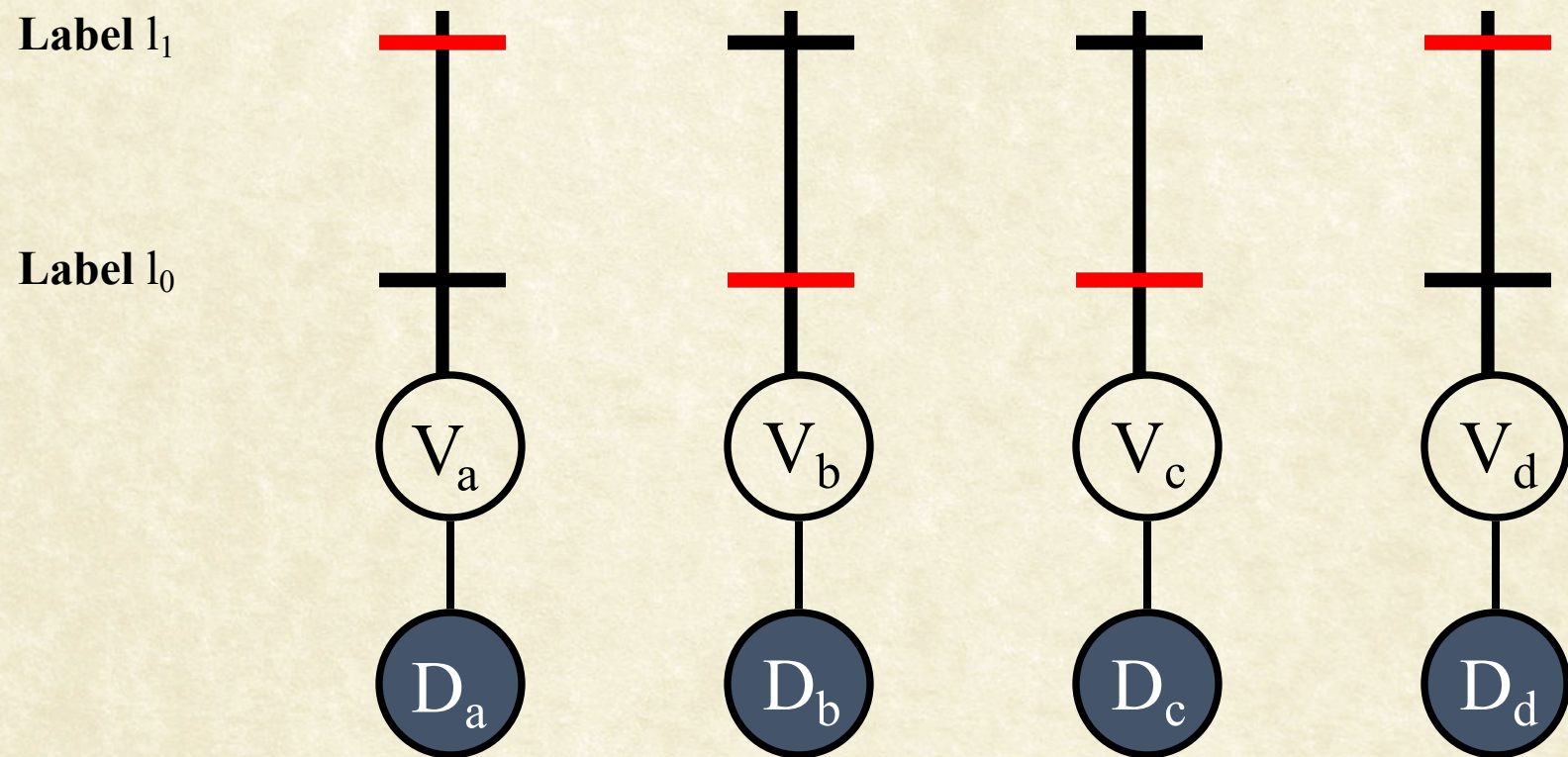
Unary Cost

Pairwise Cost

Find $f^* = \arg \min Q(f)$



Formulation: Energy Function



Random Variables $V = \{V_a, V_b, \dots\}$

Labels $L = \{l_0, l_1, \dots\}$ Data D

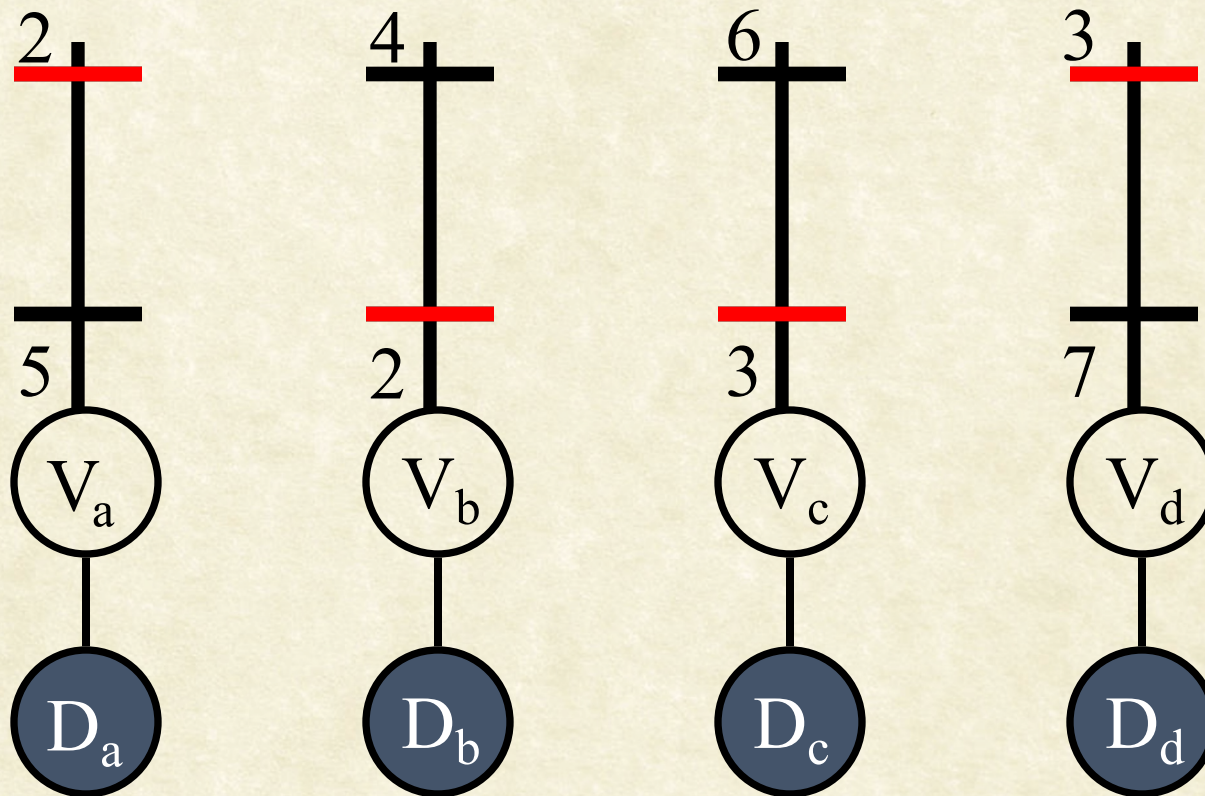
Labelling $f: \{a, b, \dots\} \rightarrow \{0, 1, \dots\}$



Energy Function

Label l_1

Label l_0



$$Q(f) = \sum_a \theta_{a;f(a)}$$

Unary Potential

Easy to minimize

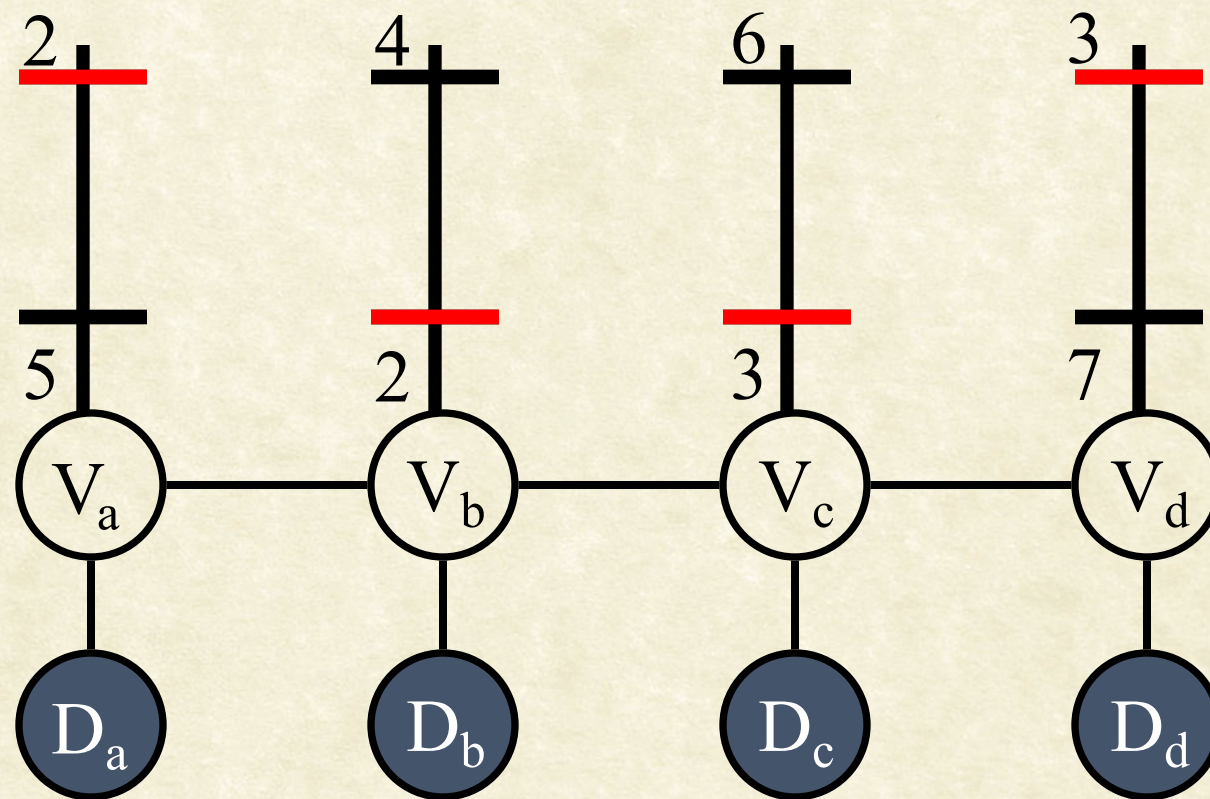
Neighbourhood



Energy Function

Label l_1

Label l_0

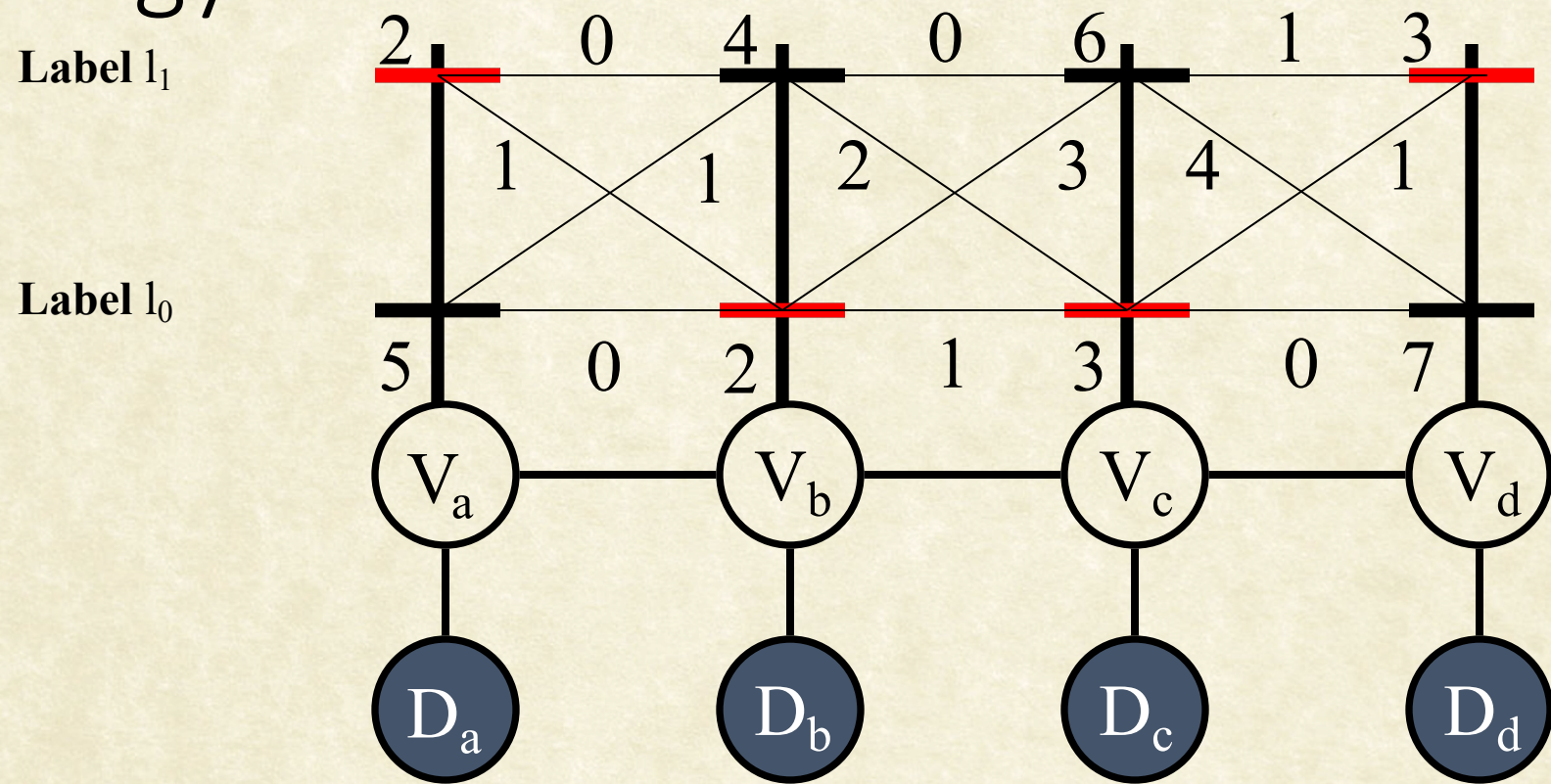


$E : (a,b) \in E$ iff V_a and V_b are neighbours

$$E = \{ (a,b) , (b,c) , (c,d) \}$$



Energy Function

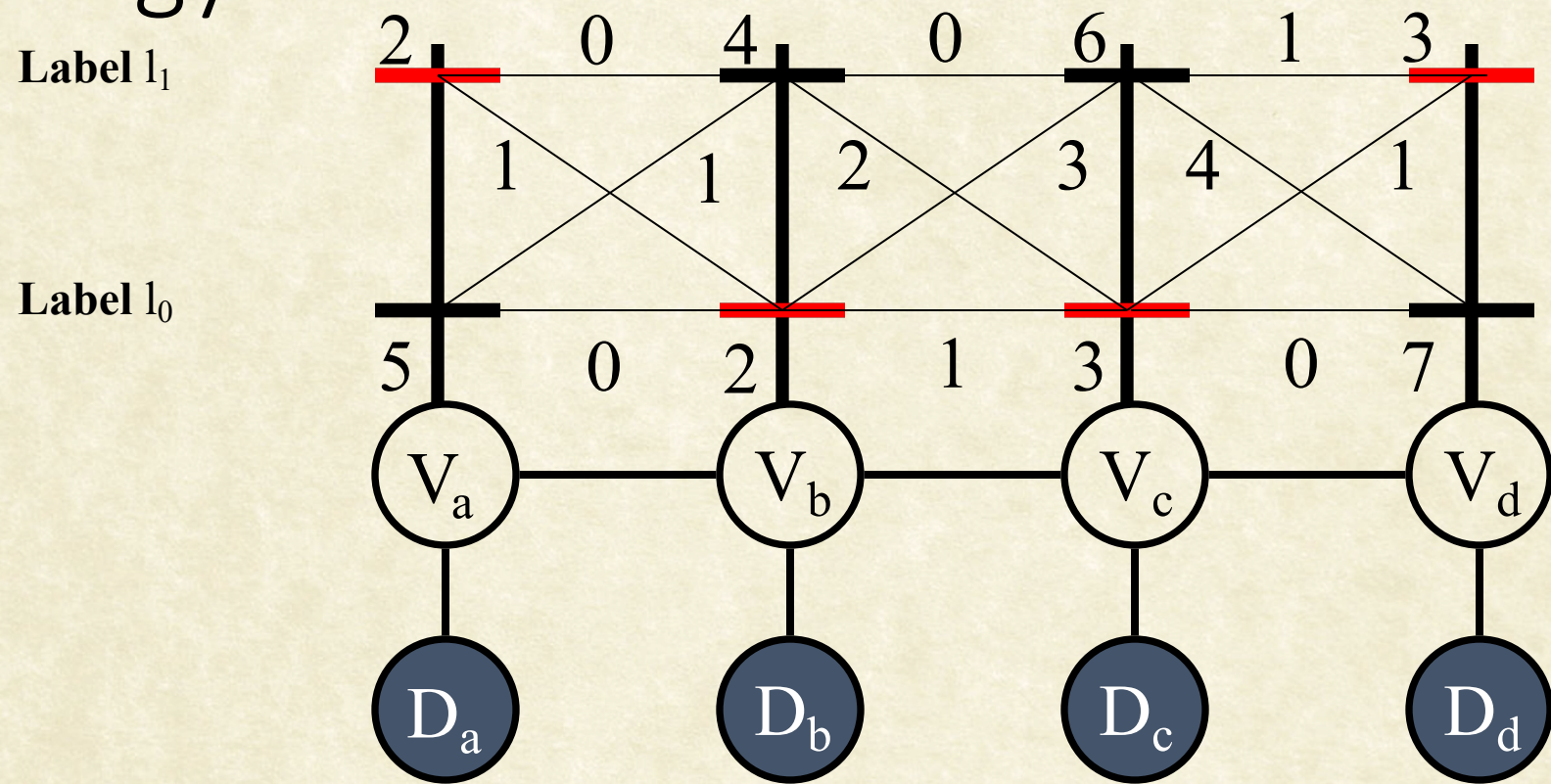


Pairwise Potential

$$Q(f) = \sum_a \theta_{a;f(a)} + \sum_{(a,b)} \theta_{ab;f(a)f(b)}$$



Energy Function

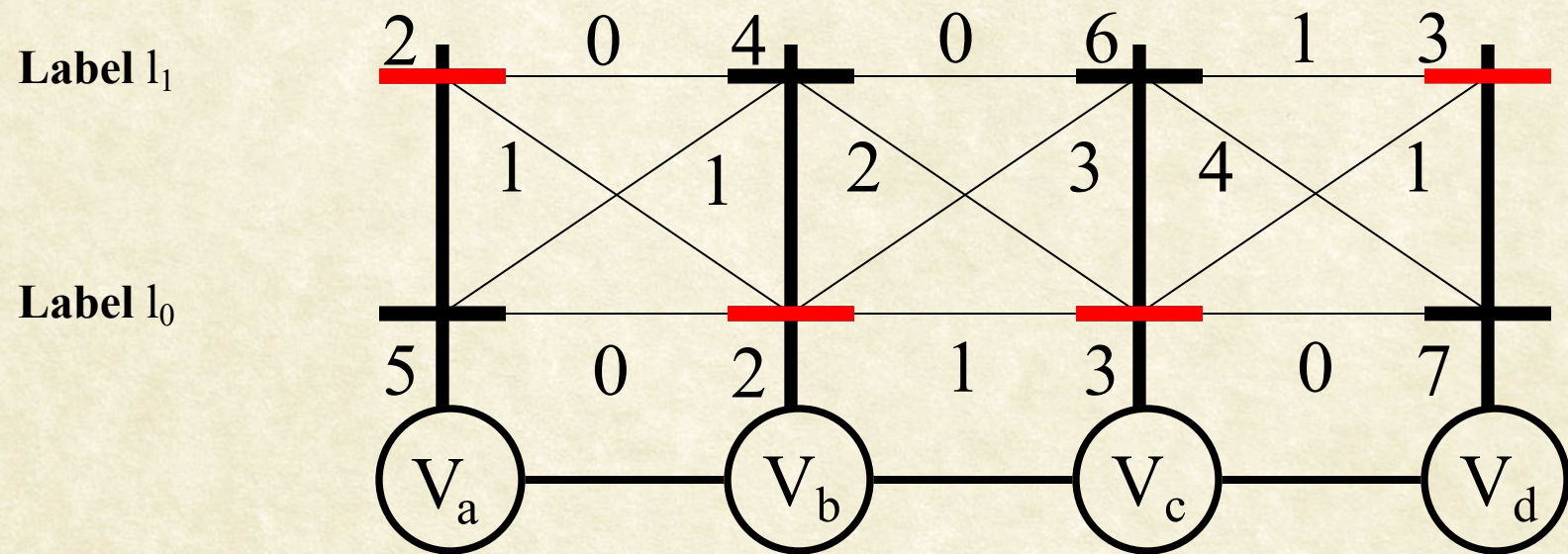


$$Q(f; \theta) = \sum_a \theta_{a;f(a)} + \sum_{(a,b)} \theta_{ab;f(a)f(b)}$$

Parameter



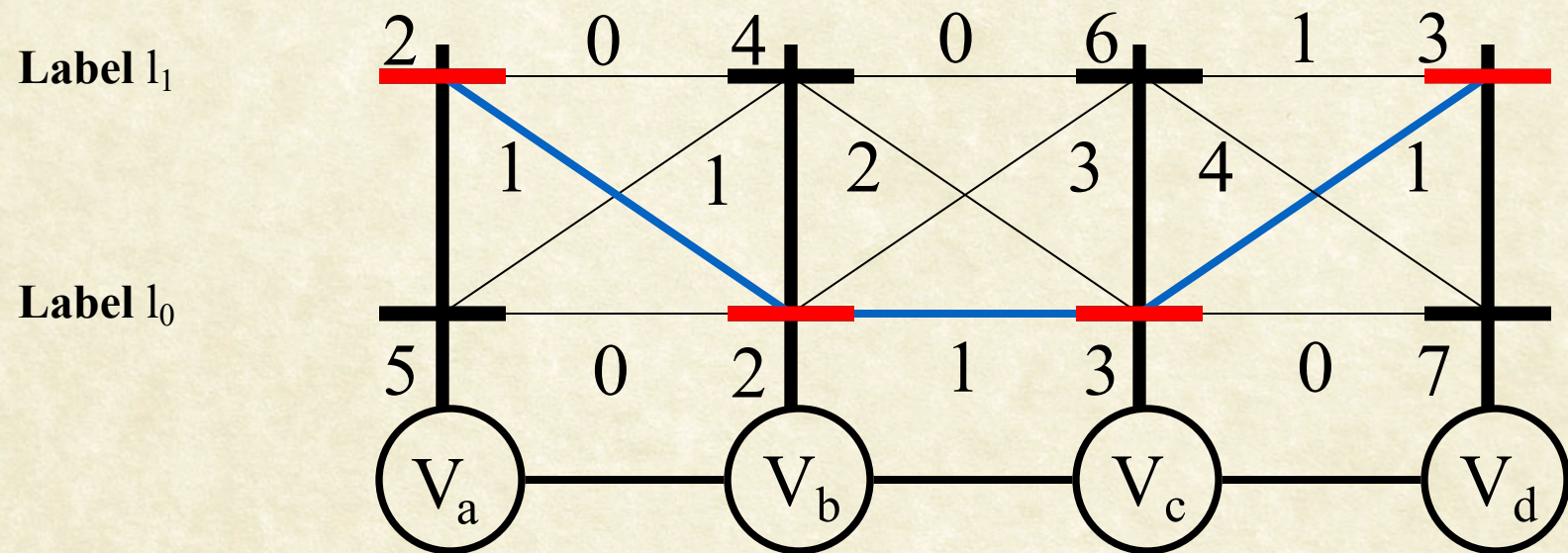
MAP Estimation



$$Q(f; \theta) = \sum_a \theta_{a;f(a)} + \sum_{(a,b)} \theta_{ab;f(a)f(b)}$$



MAP Estimation

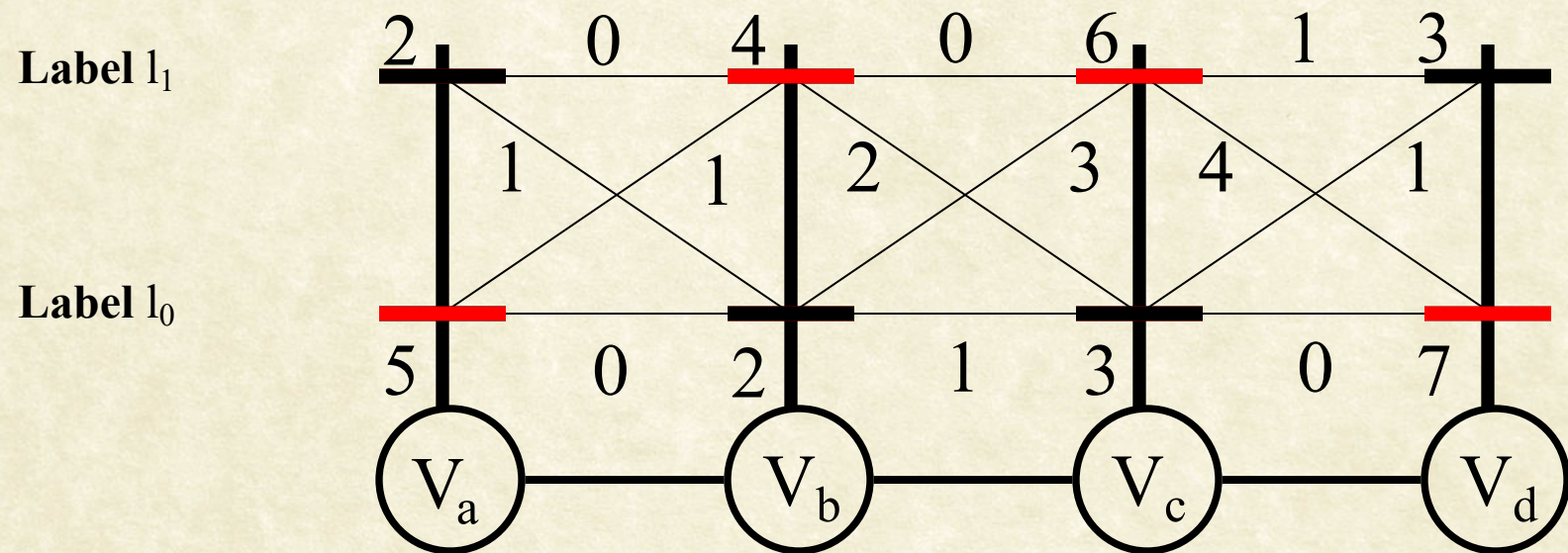


$$Q(f; \theta) = \sum_a \theta_{a;f(a)} + \sum_{(a,b)} \theta_{ab;f(a)f(b)}$$

$$2 + 1 + 2 + 1 + 3 + 1 + 3 = 13$$



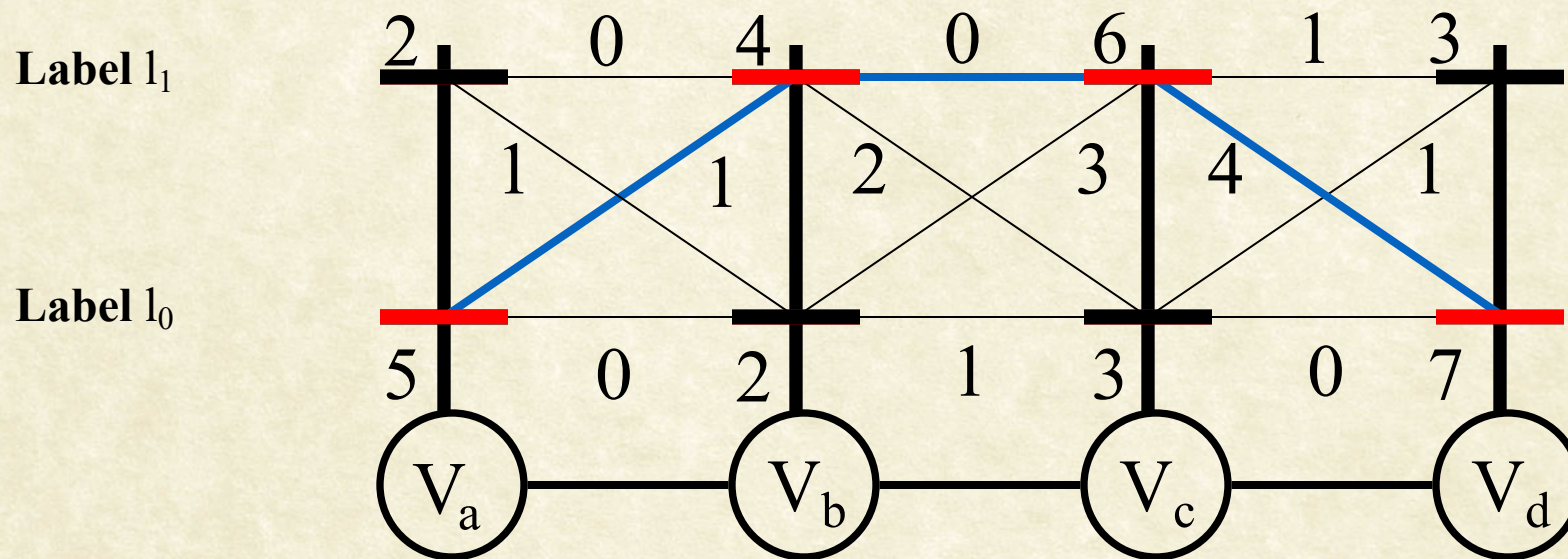
MAP Estimation



$$Q(f; \theta) = \sum_a \theta_{a;f(a)} + \sum_{(a,b)} \theta_{ab;f(a)f(b)}$$



MAP Estimation

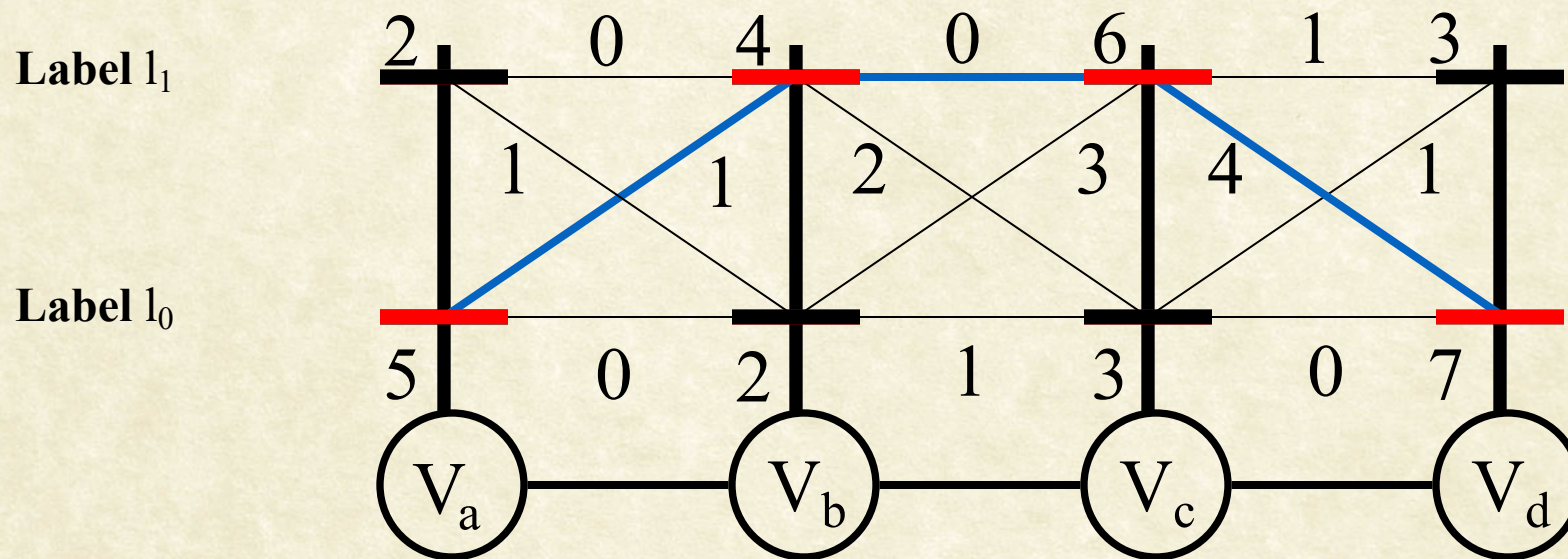


$$Q(f; \theta) = \sum_a \theta_{a;f(a)} + \sum_{(a,b)} \theta_{ab;f(a)f(b)}$$

$$5 + 1 + 4 + 0 + 6 + 4 + 7 = 27$$



MAP Estimation



$$q^* = \min Q(f; \theta) = Q(f^*; \theta)$$

$$Q(f; \theta) = \sum_a \theta_{a;f(a)} + \sum_{(a,b)} \theta_{ab;f(a)f(b)}$$

$$f^* = \arg \min Q(f; \theta)$$



MAP Estimation

16 possible labellings

$$f^* = \{1, 0, 0, 1\}$$

$$q^* = 13$$

f(a)	f(b)	f(c)	f(d)	Q(f; θ)
0	0	0	0	18
0	0	0	1	15
0	0	1	0	27
0	0	1	1	20
0	1	0	0	22
0	1	0	1	19
0	1	1	0	27
0	1	1	1	20

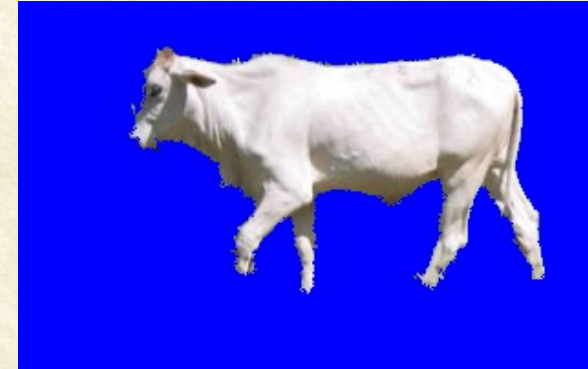
f(a)	f(b)	f(c)	f(d)	Q(f; θ)
1	0	0	0	16
1	0	0	1	13
1	0	1	0	25
1	0	1	1	18
1	1	0	0	18
1	1	0	1	15
1	1	1	0	23
1	1	1	1	16



Computational Complexity

Segmentation

$$2^{|V|}$$



$$|V| = \text{number of pixels} \approx 320 * 480 = 153600$$

$< 2^{300}$ atoms in
the universe

Can we do better than brute-force?

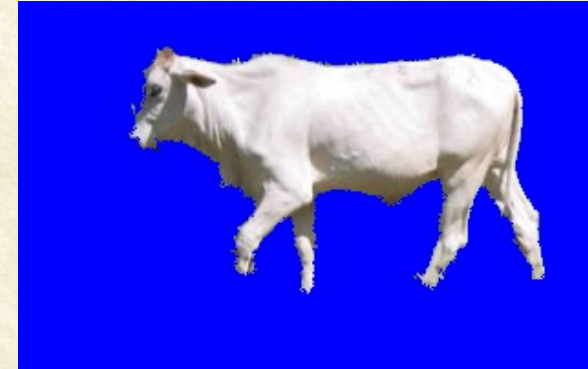
MAP Estimation is NP-hard !!



Computational Complexity

Segmentation

$$2^{|V|}$$



$$|V| = \text{number of pixels} \approx 320 * 480 = 153600$$

Exact algorithms do exist for special cases

Good approximate algorithms for general case



Further Reading

IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, VOL. 26, NO. 2, FEBRUARY 2004

147

What Energy Functions Can Be Minimized via Graph Cuts?

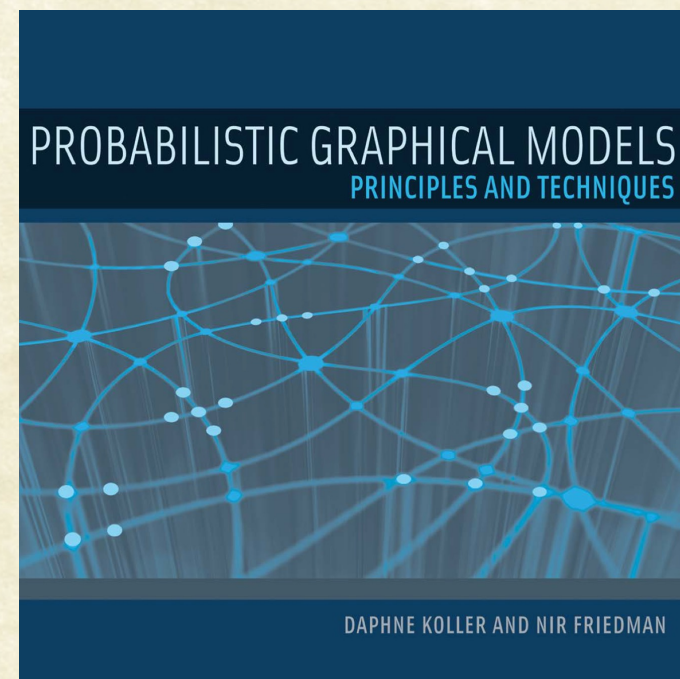
Vladimir Kolmogorov, *Member, IEEE*, and Ramin Zabih, *Member, IEEE*

1068

IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, VOL. 30, NO. 6, JUNE 2008

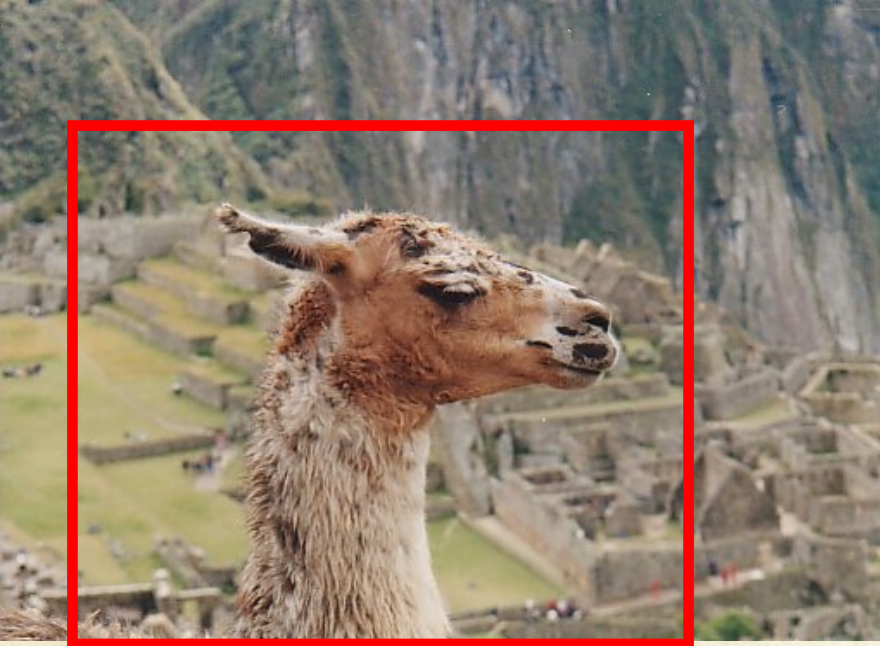
A Comparative Study of Energy Minimization Methods for Markov Random Fields with Smoothness-Based Priors

Richard Szeliski, *Fellow, IEEE*, Ramin Zabih, *Senior Member, IEEE*, Daniel Scharstein, *Member, IEEE*, Olga Veksler, *Member, IEEE*, Vladimir Kolmogorov, *Member, IEEE*, Aseem Agarwala, *Member, IEEE*, Marshall Tappen, *Member, IEEE*, and Carsten Rother, *Member, IEEE*





GrabCuts: An Intelligent Extension



Fast &
Accurate ?



- Less user input: only rectangle
- Handle color
- Extract matte as post-process



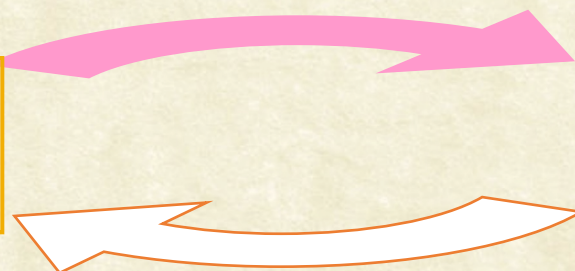
Approach: Iterated Graph Cuts



User Initialisation



**Learn foreground
color model**

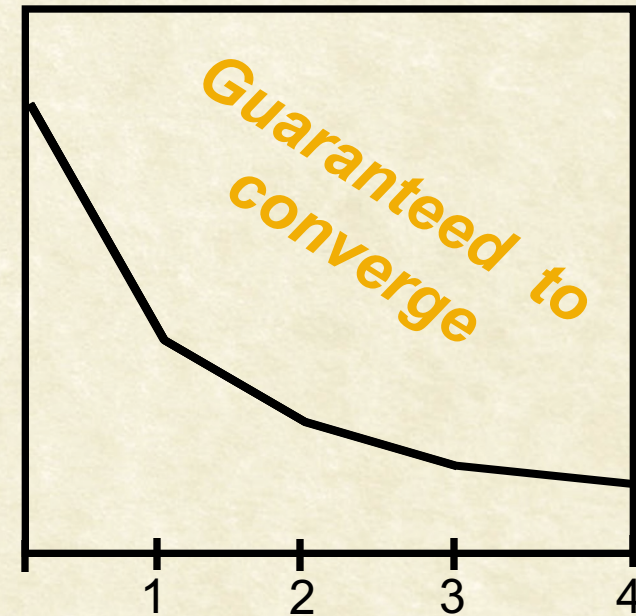


**Graph cuts to
infer the
foreground**

Iterated Graph Cuts



Result

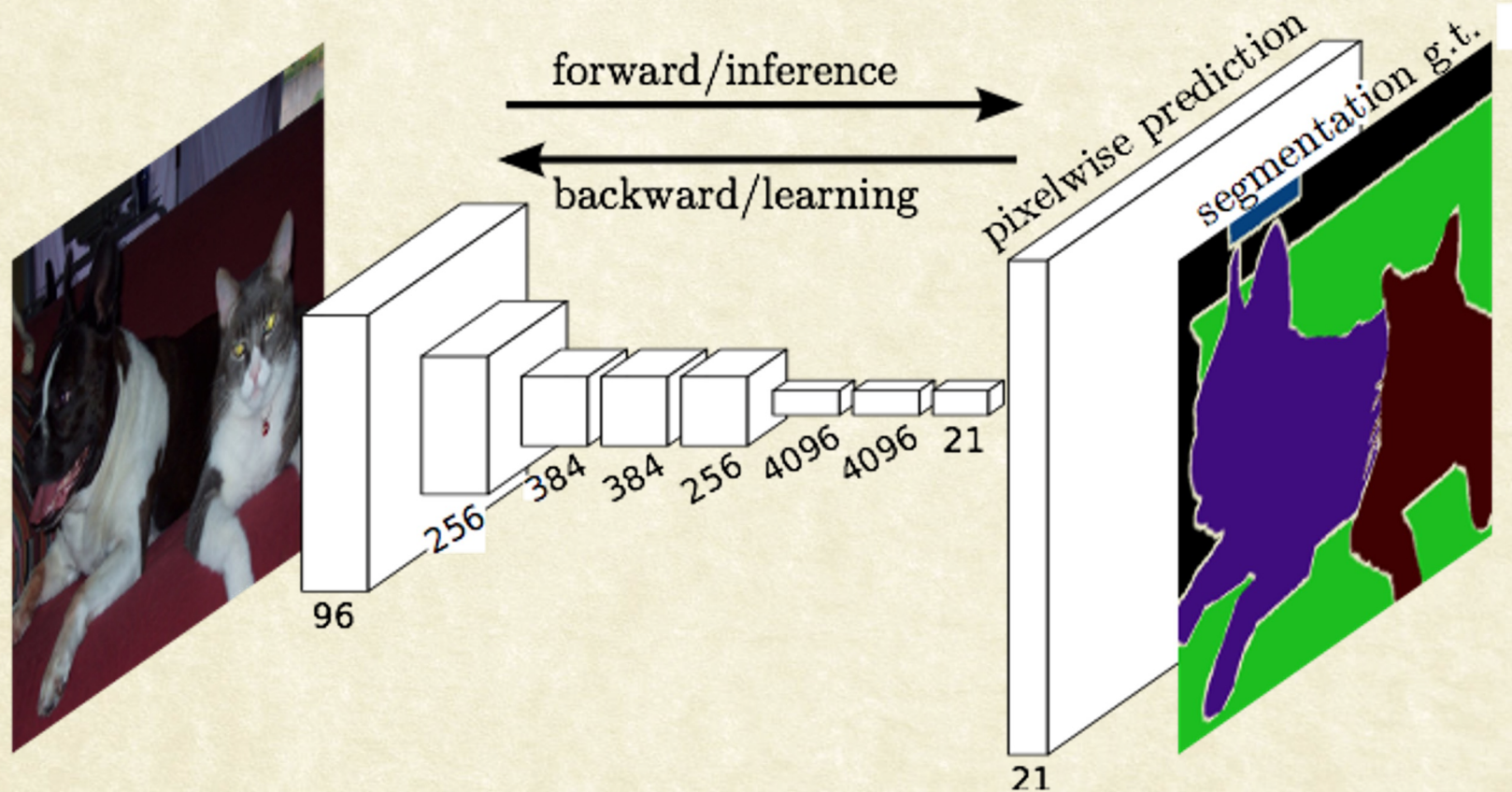


Energy after each Iteration





FCN for Semantic Segmentation



- Challenge: Segmentation = Classification + Localization
- Classification needs larger context and location invariance
- Localization needs sensitivity to location

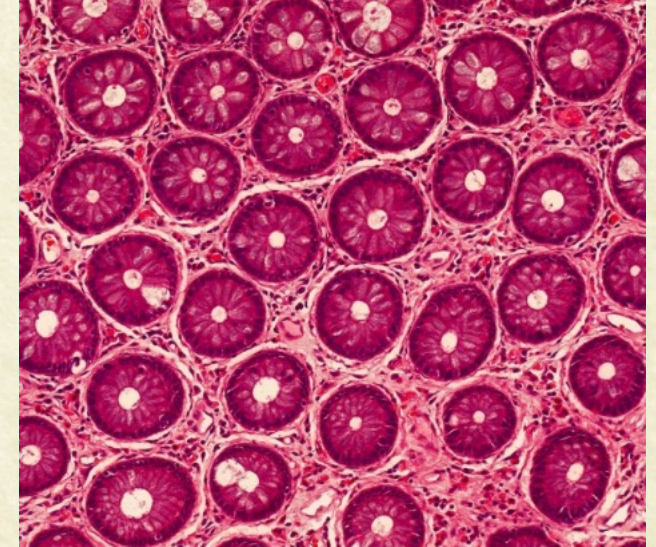
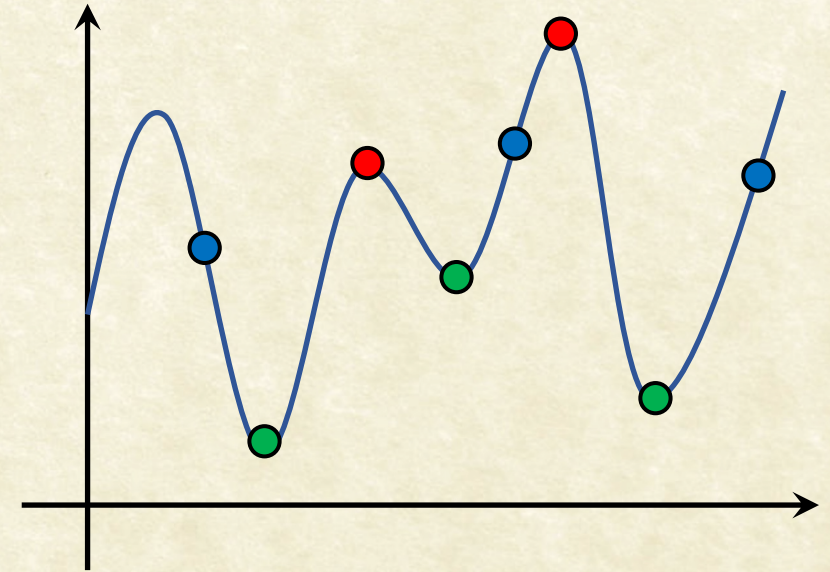


Questions?



Watershed Segmentation

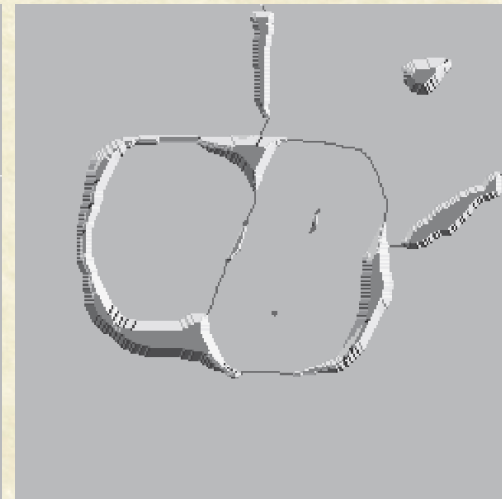
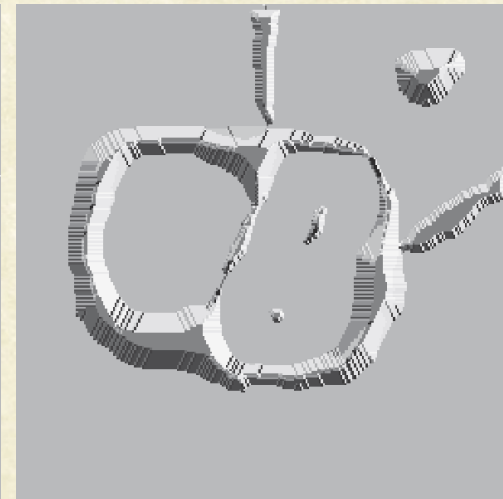
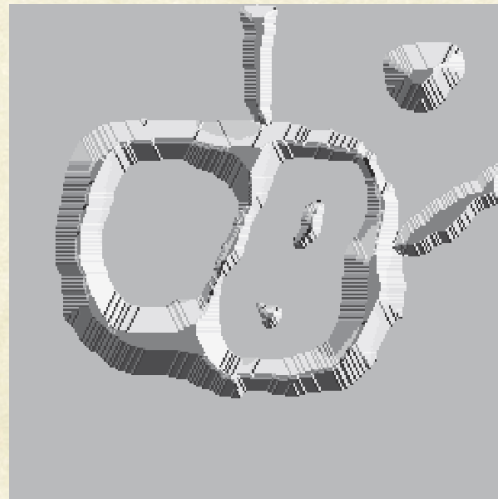
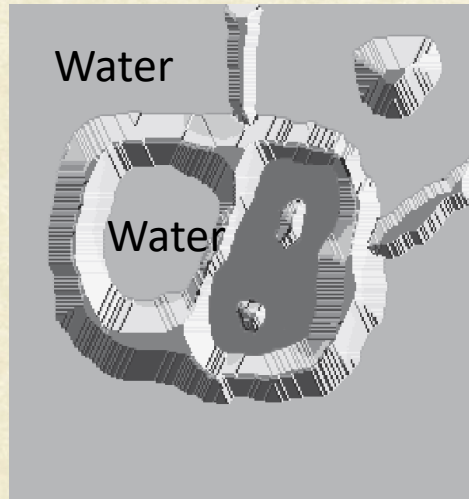
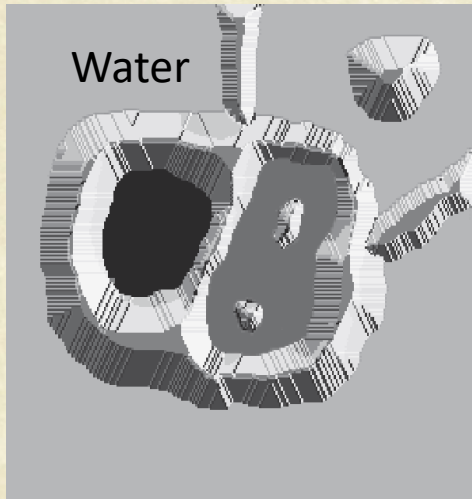
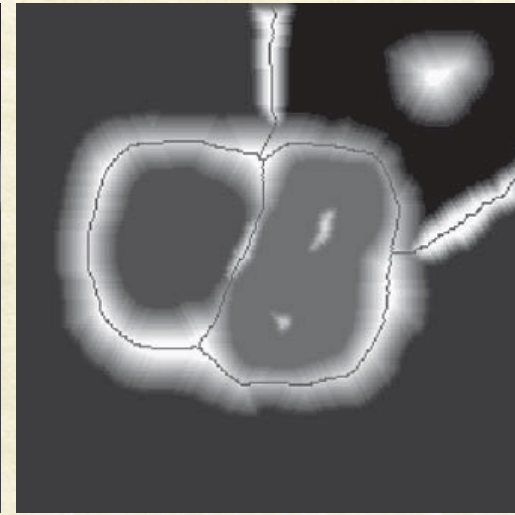
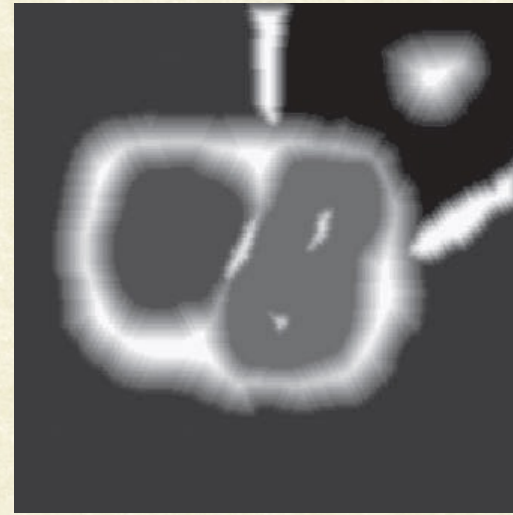
- Consider the intensity profile of an image as a topographical surface
- We can identify three types of points:
 1. Local minima, where water collects
 2. Points where water flows to a single minima (catchment basin)
 3. Points where water can flow to any one of multiple minima (watershed lines)
- Principal Goal: Find the watershed lines (object boundaries)





Watershed Segmentation

- Original Image, Topographical view
- Various stages of flooding.



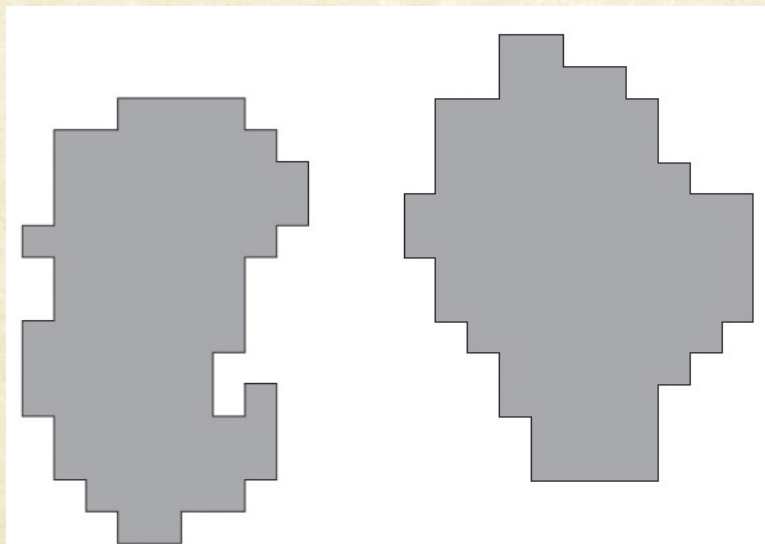


The Watershed Algorithm

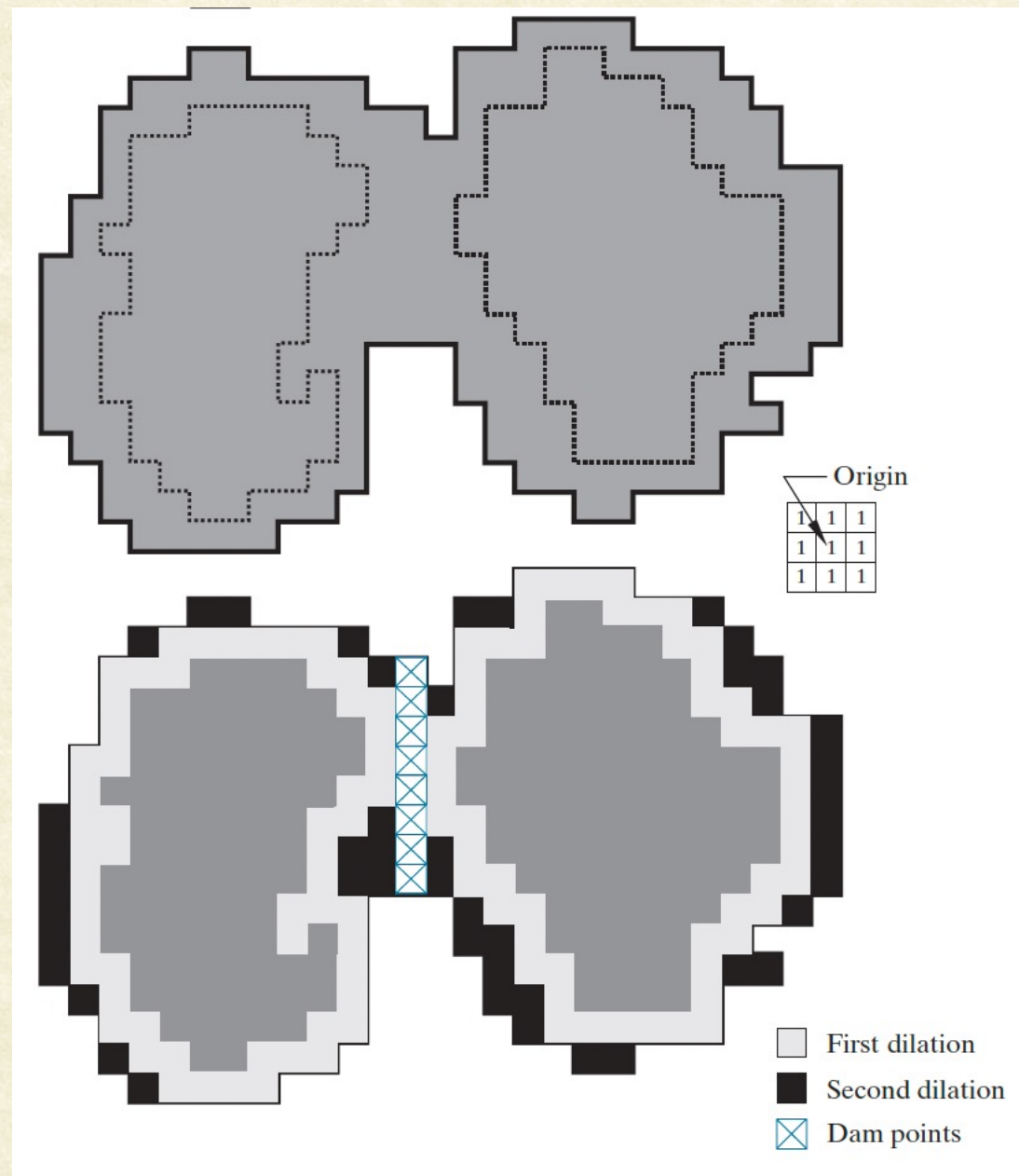
- Let $M_1, M_2, \dots, M_R$ be the regional minima.
- Let $C(M_i)$ be points in the catchment area of M_i .
- Let $T[n]$ represent the set of co-ordinates for which $g(s,t) < n$.
- Flood the topography in integer increments
 - From $n = \min + 1$ to $n = \max + 1$.
- At each stage, we compute $C_n(M_i)$ from $C_{n-1}(M_i)$ and $T[n]$ using morphological operators
 - Essentially, add pixels of value n to the corresponding $C(M_i)$



Watershed by Dilation



- Initial image and the result of two steps of dilation resulting in dam points.





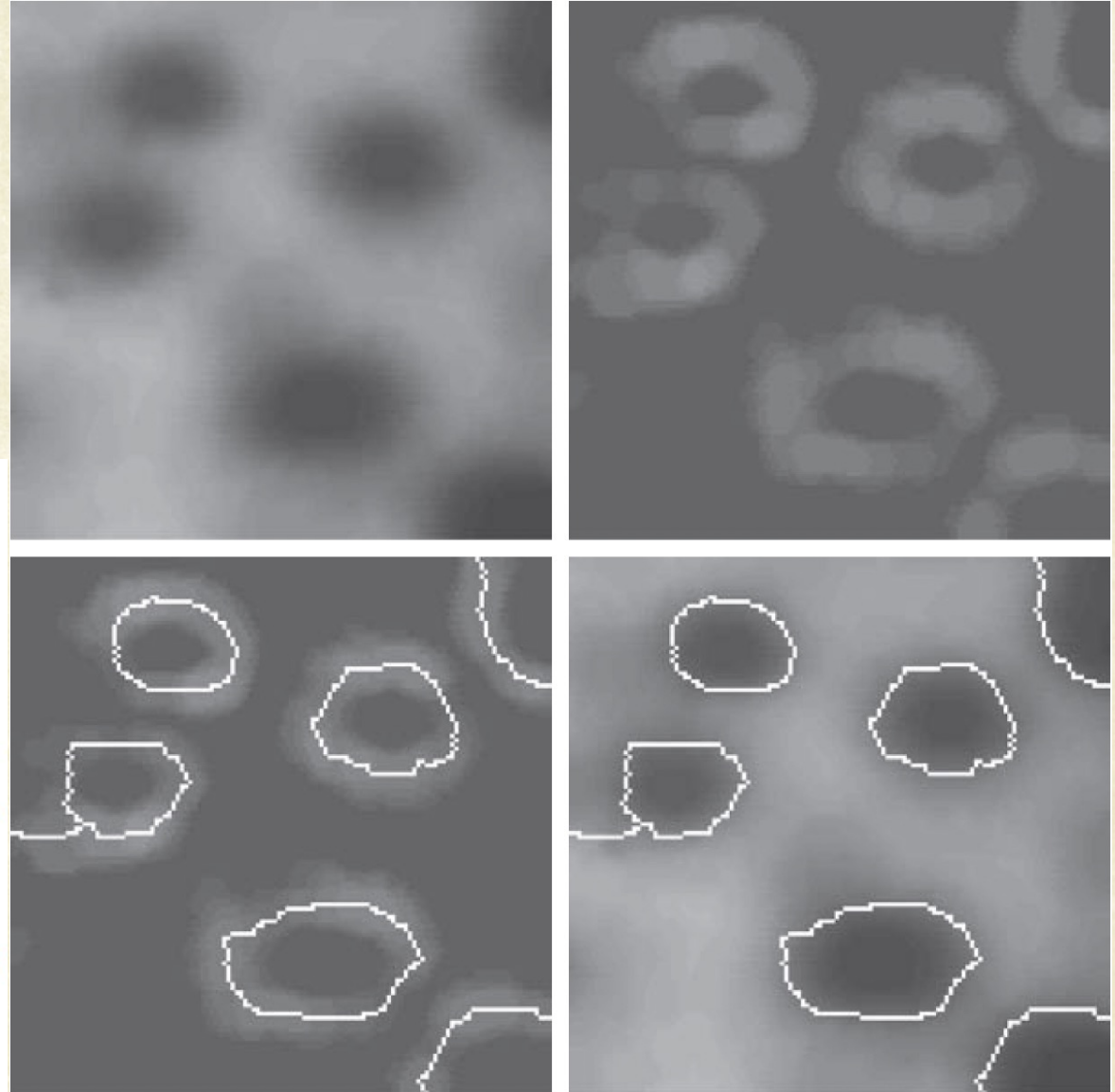
Watershed Example

- Watershed is often performed on the gradient image (top right).

a	b
c	d

FIGURE 10.59

(a) Image of blobs.
(b) Image gradient.
(c) Watershed lines, superimposed on the gradient image.
(d) Watershed lines superimposed on the original image.





Over-segmentation from Watershed

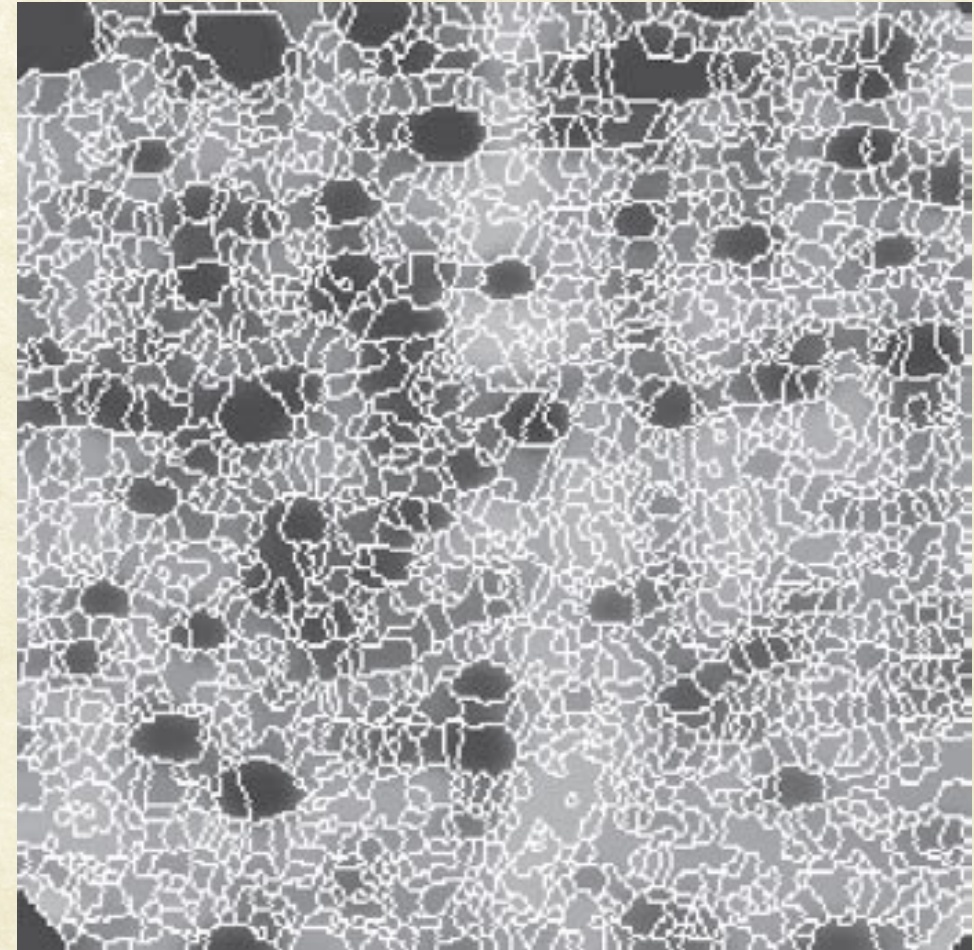
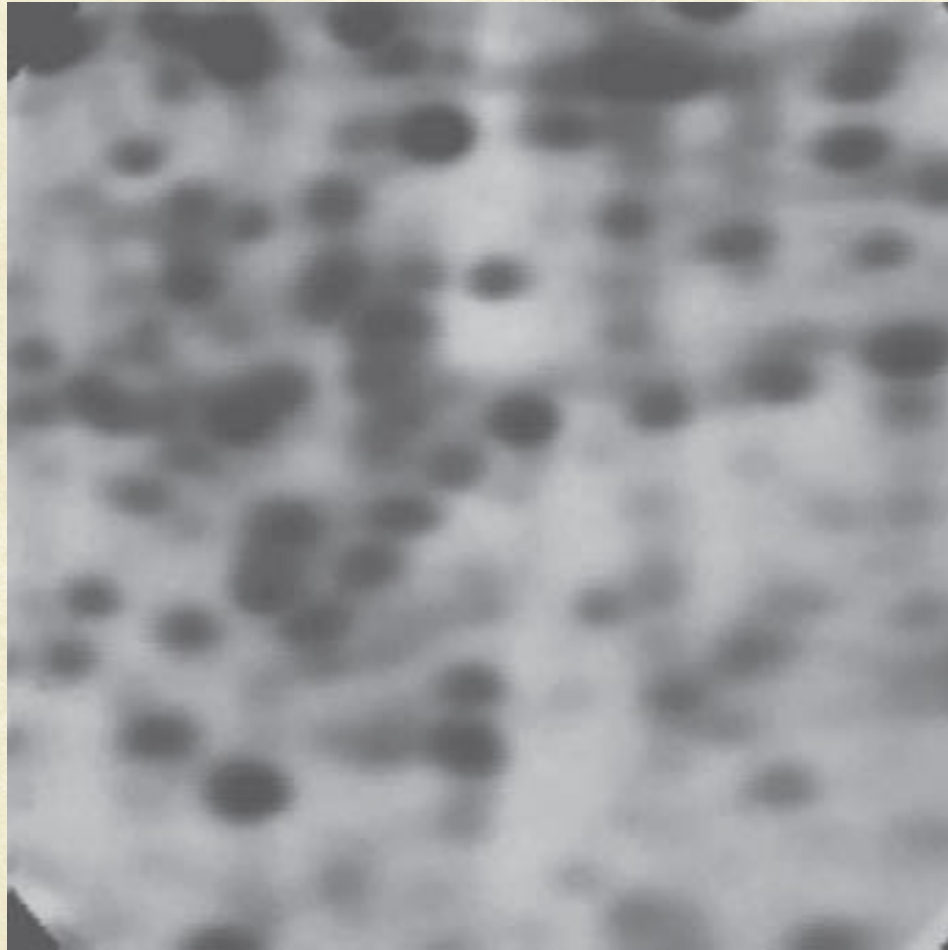
a b

FIGURE 10.60

(a) Electrophoresis image.

(b) Result of applying the watershed segmentation algorithm to the gradient image.

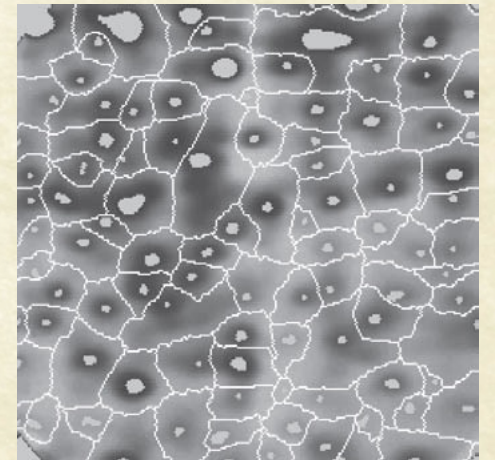
Over-segmentation is evident.





Markers to avoid Over-segmentation

- Suppose that we define an internal marker as:
 1. A region that is surrounded by points of higher “altitude”;
 2. such that the points in the region form a connected component; and
 3. in which all the points in the connected component have the same intensity value.
- Smooth the image and identify internal markers
- Use only internal markers as regional minima.



See **Section 10.7** in the textbook for details



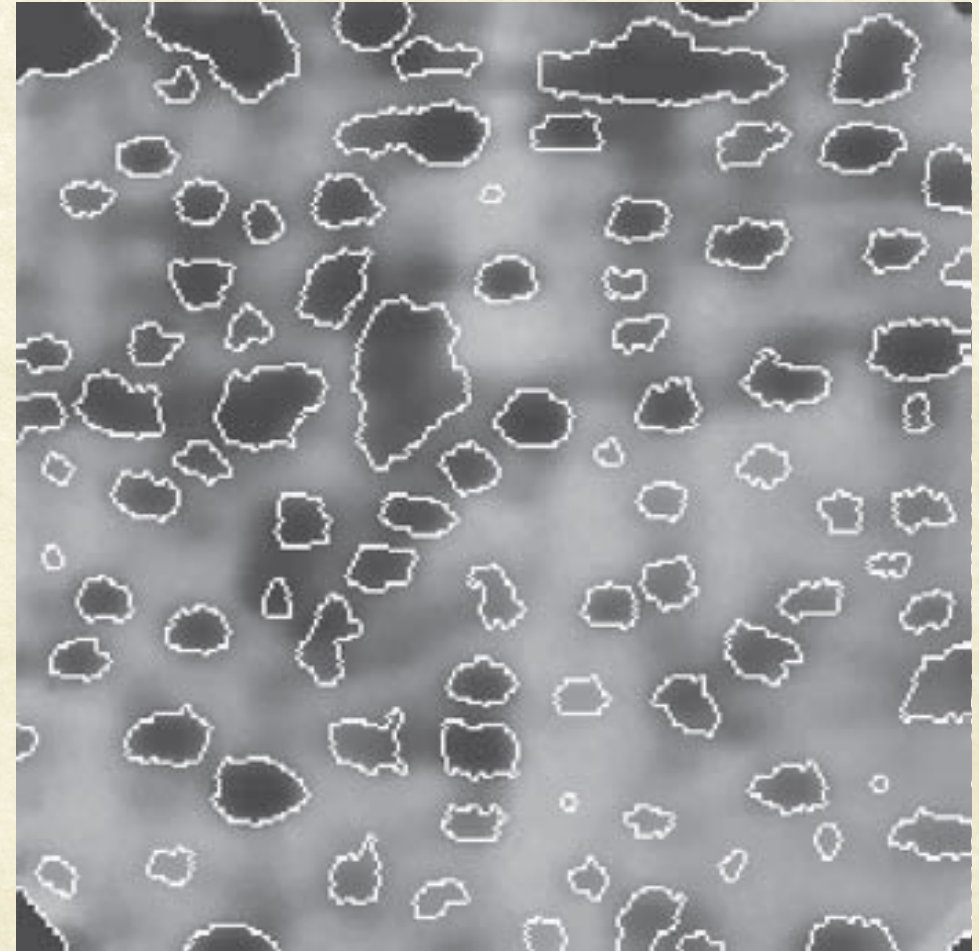
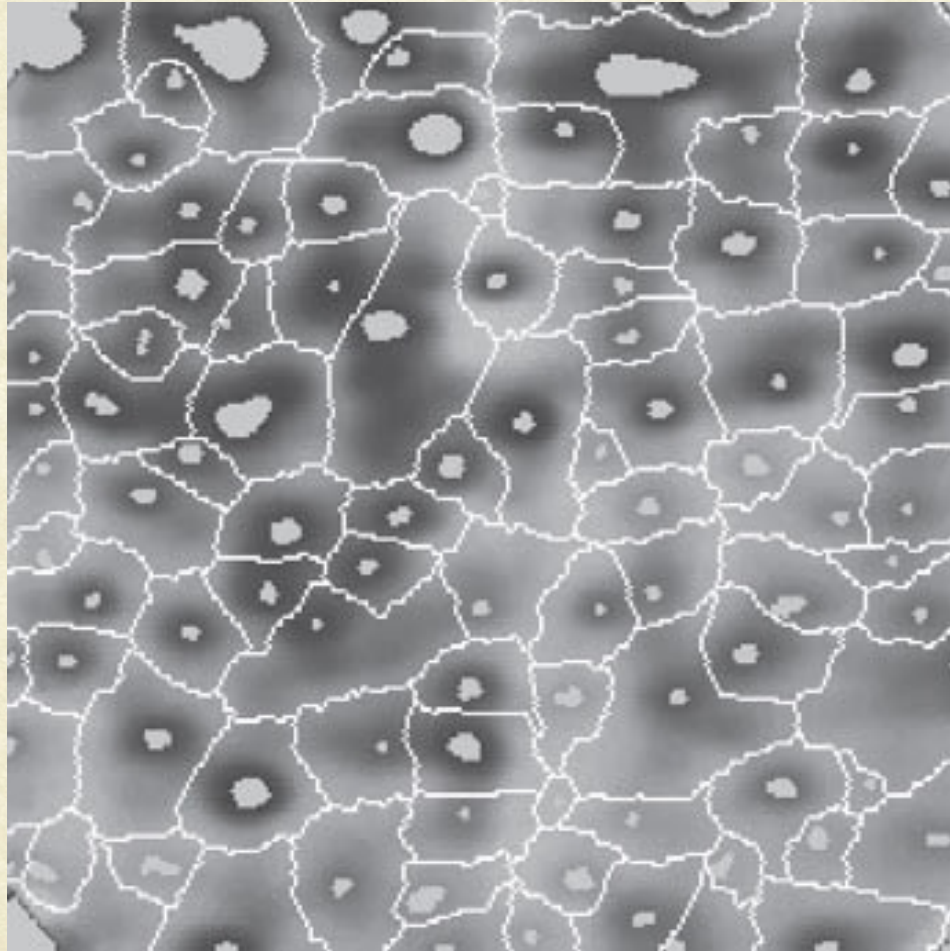
Over-segmentation from Watershed

a b

FIGURE 10.61

(a) Image showing internal markers (light gray regions) and external markers (watershed lines).

(b) Result of segmentation. Note the improvement over Fig. 10.60(b).





Questions?